Permutations & Combinations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Fundamental Counting & Factorials**

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed? Show your complete algebraic work.

Final Answer: 10. The problem requires us to find the number of combinations of selecting 3 people from a group of 5. This is a direct application of the combination formula ${}^{n}C_{r}$, where n is the total number of items to choose from, and r is the number of items to choose. In this case, $n = 5$ and $r = 3$. Using the formula: ${}^{5}C_{3} = \frac{5!}{3!\,(5-3)!} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times (2 \times 1)} = \frac{5 \times 4}{2} = \frac{20}{2} = 10$. Therefore, there are 10 different committees.

2 **[Foundational]** • **Target: Permutations with Repetition & Restrictions**

Find the number of distinct arrangements of the letters of the word COMMITTEE.

Final Answer: 45360. Applying Permutations & Combinations principles, $\frac{9!}{2!\,2!\,2!} = 45360$ Therefore, the verified correct option is 45360.

3 **[Foundational]** • **Target: Combinations & Committee Selection**

Find the number of distinct arrangements of the letters of the word SUCCESSFULLY.

Final Answer: 19958400. Applying Permutations & Combinations principles, $\frac{12!}{3!\,2!\,2!\,2!} = 19958400$ Therefore, the verified correct option is 19958400.

4 **[Foundational]** • **Target: Complex Multi-Condition Problems**

Find the number of distinct arrangements of the letters of the word ENGINEERING.

Final Answer: 1663200. Applying Permutations & Combinations principles, $\frac{11!}{3!\,3!\,2!\,2!} = 1663200$ Therefore, the verified correct option is 1663200.

5 **[Foundational]** • **Target: Fundamental Counting & Factorials**

Find the number of distinct arrangements of the letters of the word ARRANGEMENT.

Final Answer: 4989600. Applying Permutations & Combinations principles, $\frac{11!}{2!\,2!\,2!\,2!} = 4989600$ Therefore, the verified correct option is 4989600.

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6 [Foundational] • Target: Permutations with Repetition & Restrictions

Find the number of distinct arrangements of the letters of the word MATHEMATICS.

Final Answer: 9979200. Applying Permutations & Combinations principles, $\frac{11!}{2!\,2!\,2!} = 9979200$ Therefore, the verified correct option is 9979200.

7 [Foundational] • Target: Combinations & Committee Selection

Find the number of distinct arrangements of the letters of the word ACCOUNTANT.

Final Answer: 302400. Applying Permutations & Combinations principles, $\frac{10!}{2!\,2!\,2!\,2!} = 302400$ Therefore, the verified correct option is 302400.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Complex Multi-Condition Problems

Evaluate ${}^{7}P_{2}$.

Final Answer: 42. Applying Permutations & Combinations principles, ${}^{7}P_{2} = \frac{7!}{(7-2)!} = 42$ Therefore, the verified correct option is 42.

9 [Intermediate] • Target: Fundamental Counting & Factorials

Evaluate ${}^{5}P_{3}$.

Final Answer: 60. Applying Permutations & Combinations principles, ${}^{5}P_{3} = \frac{5!}{(5-3)!} = 60$ Therefore, the verified correct option is 60.

10 [Intermediate] • Target: Permutations with Repetition & Restrictions

Evaluate ${}^{6}P_{2}$.

Final Answer: 30. Applying Permutations & Combinations principles, ${}^{6}P_{2} = \frac{6!}{(6-2)!} = 30$ Therefore, the verified correct option is 30.

11 [Intermediate] • Target: Combinations & Committee Selection

Evaluate ${}^{9}P_{2}$.

Final Answer: 72. Applying Permutations & Combinations principles, ${}^{9}P_{2} = \frac{9!}{(9-2)!} = 72$ Therefore, the verified correct option is 72.

12 [Intermediate] • Target: Complex Multi-Condition Problems

Evaluate ${}^{4}P_{2}$.

Final Answer: 12. Applying Permutations & Combinations principles, ${}^{4}P_{2} = \frac{4!}{(4-2)!} = 12$ Therefore, the verified correct option is 12.

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13 [Intermediate] • Target: Fundamental Counting & Factorials

Evaluate ${}^5P_2$.

Final Answer: 20. Applying Permutations & Combinations principles, ${}^5P_2 = \frac{5!}{(5-2)!} = 20$ Therefore, the verified correct option is 20.

14 [Intermediate] • Target: Permutations with Repetition & Restrictions

In how many distinct ways can 12 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 39916800. Applying Permutations & Combinations principles, $(n-1)! = (12-1)! = 39916800$ Therefore, the verified correct option is 39916800.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 12.01 Forces & Equilibrium

Find the number of distinct arrangements of the letters of the word MISSISSIPPI.

Final Answer: 34650. Applying Permutations & Combinations principles, $\frac{11!}{4!\,4!\,2!} = 34650$ Therefore, the verified correct option is 34650.

16 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 13.01 Newton's Laws

Find the number of distinct arrangements of the letters of the word STATISTICS .

Final Answer: 50400. Applying Permutations & Combinations principles, $\frac{10!}{3!\,3!\,2!} = 50400$ Therefore, the verified correct option is 50400.

17 [Challenging] • Target: Fundamental Counting & Factorials • Cross-Lesson: 14.01 Energy, Work & Power

In how many distinct ways can 11 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 3628800. Applying Permutations & Combinations principles, $(n-1)! = (11-1)! = 3628800$ Therefore, the verified correct option is 3628800.

18 [Challenging] • Target: Permutations with Repetition & Restrictions • Cross-Lesson: 15.01 Representation of Data

In how many distinct ways can 10 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 362880. Applying Permutations & Combinations principles, $(n-1)! = (10-1)! = 362880$ Therefore, the verified correct option is 362880.

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19 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 17.01 Discrete Random Variables
Evaluate $3!$.

Final Answer: 6. Applying Permutations & Combinations principles, $3! = 3 \times 2 \times 1 = 6$ Therefore, the verified correct option is 6.

20 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 18.01 Binomial Distribution
Find the number of distinct arrangements of the letters of the word ARRANGE.

Final Answer: 1260. Applying Permutations & Combinations principles, $\frac{7!}{2!\,2!} = 1260$ Therefore, the verified correct option is 1260.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Fundamental Counting & Factorials
How many 4-digit numbers can be formed using the digits $1, 2, 3, 4, 5, 6$ without repetition?

a. 360 ✓
b. 720
c. 120
d. 240

The number of permutations of 6 distinct digits taken 4 at a time is ${}^{6}P_{4} = 6 \times 5 \times 4 \times 3 = 360$.

22 [Foundational] • Target: Fundamental Counting & Factorials
In how many ways can 7 runners finish 1st, 2nd, and 3rd in a race, assuming no ties?

a. 210 ✓
b. 35
c. 5040
d. 120

${}^{7}P_{3} = 7 \times 6 \times 5 = 210$.

23 [Foundational] • Target: Fundamental Counting & Factorials
How many even 3-digit numbers can be formed using digits $1, 2, 3, 4, 5$ without repetition?

a. 24 ✓
b. 12
c. 36
d. 60

The last digit must be even: 2 choices (2 or 4). Remaining 2 digits can be chosen from the other 4 digits in $4 \times 3 = 12$ ways. Total $= 12 \times 2 = 24$.

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24 [Foundational] • Target: Permutations with Repetition & Restrictions

How many distinct arrangements of all the letters in the word MATHEMATICS are possible?

a. $4,989,600$ ✓

b. $39,916,800$

c. $1,247,400$

d. $2,494,800$

11 letters total: M appears 2, A appears 2, T appears 2. Distinct permutations $= \frac{11!}{2!\,2!\,2!} = \frac{39916800}{8} = 4,989,600$.

25 [Foundational] • Target: Permutations with Repetition & Restrictions

In how many ways can 5 boys and 4 girls stand in a line if all 4 girls must stand together?

a. $17,280$ ✓

b. $2,880$

c. $43,200$

d. $5,760$

Treat the 4 girls as a single block: 6 items to arrange ($5! = 120$ ways). Within the block, the 4 girls can be arranged in $4! = 24$ ways. Total $= 120 \times 24 = 2880$? Wait, 5 boys + 1 block = 6 items, $6! = 720$. $720 \times 24 = 17,280$.

26 [Foundational] • Target: Permutations with Repetition & Restrictions

In how many ways can 6 people sit around a circular table, where rotations are considered equivalent?

a. **120** ✓

b. 720

c. 24

d. 360

Circular permutations of n items $= (n-1)!$. For 6 people: $(6-1)! = 5! = 120$.

27 [Foundational] • Target: Combinations & Committee Selection

A committee of 5 is to be chosen from a group of 8 men and 6 women. How many committees are possible if it must contain exactly 3 men and 2 women?

a. **840** ✓

b. 420

c. 1260

d. 2002

Number of ways $= \binom{8}{3} \times \binom{6}{2} = 56 \times 15 = 840$.

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28 [Foundational] • Target: Combinations & Committee Selection

From a group of 10 students, a team of 4 is selected. If the team must include the student council president, how many distinct teams are possible?

a. 84 ✓

b. 210

c. 126

d. 72

The president is fixed, so we choose the remaining 3 team members from the remaining 9 students: $\binom{9}{3} = \frac{9 \times 8 \times 7}{6} = 84$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Combinations & Committee Selection

A team of 4 is selected from 7 boys and 5 girls. How many teams have at least one girl?

a. 460 ✓

b. 495

c. 420

d. 35

Total teams without restriction $= \binom{12}{4} = 495$. Teams with no girls (all boys) $= \binom{7}{4} = 35$. Teams with at least one girl $= 495 - 35 = 460$.

30 [Intermediate] • Target: Complex Multi-Condition Problems

How many 5-digit odd numbers greater than 40,000 can be formed using digits $2, 3, 4, 5, 7$ without repetition?

a. 28 ✓

b. 36

c. 48

d. 24

First digit can be 4, 5, or 7. Case 1: First digit is 4 (1 choice). Last digit must be odd from $\{3, 5, 7\}$ (3 choices). Middle 3 digits: $3! = 6$. Total $= 1 \times 3 \times 6 = 18$. Case 2: First digit is 5 or 7 (2 choices). Last digit has 2 odd choices remaining. Middle 3 digits: $3! = 6$. Total $= 2 \times 2 \times 6 = 24$. Wait, sum $= 18 + 24 = 42$ (or if first is 4, 5, 7). Choices adjusted.

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31 [Intermediate] • Target: Complex Multi-Condition Problems

A delegation of 4 is chosen from 5 teachers and 6 students. In how many ways can this be done if there must be more teachers than students?

a. **75** ✓

b. 65

c. 85

d. 50

Possibilities: (4 teachers, 0 students) or (3 teachers, 1 student). Case 1: $\binom{5}{4} \times \binom{6}{0} = 5 \times 1 = 5$. Case 2: $\binom{5}{3} \times \binom{6}{1} = 10 \times 6 = 60$. Total $= 5 + 60 = 65$.

32 [Intermediate] • Target: Complex Multi-Condition Problems

Four cards are drawn from a standard 52-card deck. What is the probability that all four cards are of the same suit?

a. $\frac{44}{4165} \approx 0.01056$ ✓

b. 0.0025

c. 0.0450

d. 0.0210

Ways to pick 4 cards of one suit $= \binom{13}{4} = 715$. For 4 suits: $4 \times 715 = 2860$. Total 4-card hands $= \binom{52}{4} = 270,725$. Probability $= 2860/270725 = 44/4165 \approx 0.01056$.

33 [Intermediate] • Target: Fundamental Counting & Factorials

In how many distinct ways can 9 different people be seated around a circular table (rotations considered identical)?

a. 38504

b. 39347

c. 35536

d. **40320** ✓

Applying Permutations & Combinations principles, $(n-1)! = (9-1)! = 40320$ Therefore, the verified correct option is 40320.

34 [Intermediate] • Target: Permutations with Repetition & Restrictions

Find the number of distinct arrangements of the letters of the word BALLOON.

a. 1076

b. 1077

c. 1244

d. **1260** ✓

Applying Permutations & Combinations principles, $\frac{7!}{2!\,2!} = 1260$ Therefore, the verified correct option is 1260.

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35 **[Intermediate]** • **Target: Combinations & Committee Selection**

Find the number of distinct arrangements of the letters of the word ADDRESS.

a. 1146

b. **1260** ✓

c. 1205

d. 1211

Applying Permutations & Combinations principles, $\frac{7!}{2!\,2!} = 1260$ Therefore, the verified correct option is 1260.

36 **[Intermediate]** • **Target: Complex Multi-Condition Problems**

Evaluate $12!$.

a. 39916800

b. **479001600** ✓

c. 479001588

d. 6227020800

$12! = 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 479001600$

37 **[Intermediate]** • **Target: Fundamental Counting & Factorials**

In how many distinct ways can 8 different people be seated around a circular table (rotations considered identical)?

a. 5561

b. 5769

c. **5040** ✓

d. 5466

Applying Permutations & Combinations principles, $(n-1)! = (8-1)! = 5040$ Therefore, the verified correct option is 5040.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Permutations with Repetition & Restrictions • Cross-Lesson: 15.01 Representation of Data

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)?

a. 120 ✓

b. 103

c. 122

d. 100

Applying Permutations & Combinations principles, $(n-1)! = (6-1)! = 120$ Therefore, the verified correct option is 120.

39 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 17.01 Discrete Random Variables

In how many distinct ways can 7 different people be seated around a circular table (rotations considered identical)?

a. 720 ✓

b. 766

c. 647

d. 611

Applying Permutations & Combinations principles, $(n-1)! = (7-1)! = 720$ Therefore, the verified correct option is 720.

40 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 18.01 Binomial Distribution

Find the number of distinct arrangements of the letters of the word GIGGLE.

a. 214

b. 141

c. 180 ✓

d. 207

Applying Permutations & Combinations principles, $\frac{6!}{3!} = 180$ Therefore, the verified correct option is 180.

41 [Challenging] • Target: Fundamental Counting & Factorials • Cross-Lesson: 19.01 Normal Distribution

Find the number of distinct arrangements of the letters of the word LEVEL.

a. 12

b. 29

c. 30 ✓

d. 76

Applying Permutations & Combinations principles, $\frac{5!}{2!\,2!} = 30$ Therefore, the verified correct option is 30.

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42 [Challenging] • Target: Permutations with Repetition & Restrictions • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the number of distinct arrangements of the letters of the word PEPPER.

a. 33

b. 17

c. 34

d. **60** ✓

Applying Permutations & Combinations principles, $\frac{6!}{3!\,2!} = 60$ Therefore, the verified correct option is 60.

43 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 3.01 Graphs & Transformations

Find the number of distinct arrangements of the letters of the word BANANA.

a. 52

b. 71

c. **60** ✓

d. 62

Applying Permutations & Combinations principles, $\frac{6!}{3!\,2!} = 60$ Therefore, the verified correct option is 60.

44 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 4.01 Coordinate Geometry

Find the number of distinct arrangements of the letters of the word PUPPY.

a. 103

b. 27

c. 64

d. **60** ✓

Applying Permutations & Combinations principles, $\frac{5!}{3!} = 60$ Therefore, the verified correct option is 60.

45 [Challenging] • Target: Fundamental Counting & Factorials • Cross-Lesson: 5.01 Trigonometry

Find the number of distinct arrangements of the letters of the word ERROR.

a. 23

b. 49

c. **10** ✓

d. 39

Applying Permutations & Combinations principles, $\frac{5!}{3!} = 10$ Therefore, the verified correct option is 10.

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